Transport–Spectral RH Program: Established Spine, Conditional Bridge, and Exact Missing Tasks

Abstract

This document consolidates the current rigorous spine of the transport–spectral contradiction program aimed at the Riemann Hypothesis. The established parts are: (i) Gaussian self-duality, theta symmetry, and the functional equation; (ii) the excess-transport formalism; (iii) vanishing excess under an exact window law; (iv) positive excess from an off-critical zero under a target bridge inequality. The resulting contradiction yields a clean conditional RH theorem. The remaining open part is isolated precisely as the transport-to-bank bridge.

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1 Status summary

The intended contradiction has the form

$$E_{\mathrm{ex}}(f) = 0 \qquad \text{and} \qquad \exists \rho \text{ off-critical} \Longrightarrow E_{\mathrm{ex}}(f) > 0.$$

If both statements are proved unconditionally for the same residue channel f, then the Riemann Hypothesis follows.

At the present stage, the clean situation is:

S1. The closure side has a usable standalone form: an exact window law implies $E_{\mathrm{ex}}(f) = 0$.

S2. The obstruction side has the correct coercive target: an off-critical zero should force a lower bound through an unbounded lifted drift penalty.

S3. The unresolved point is the rigorous bridge from the off-critical spectral component to the bank-level lower bound.

2 Gaussian self-duality and the functional equation

2.1 Gaussian self-duality

Lemma 2.1 (Gaussian self-duality)**.** *Let*

$$g(x) := e^{-\pi x^2}, \qquad \widehat{f}(\xi) := \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi}\, dx.$$

Then

$$\widehat{g}(\xi) = e^{-\pi \xi^2}.$$

Proof. Define

$$F(\xi) := \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x \xi}\, dx.$$

Differentiate under the integral:

$$F'(\xi) = \int_{-\infty}^{\infty} e^{-\pi x^2} (-2\pi i x) e^{-2\pi i x \xi}\, dx.$$

Since

$$\frac{d}{dx} e^{-\pi x^2} = -2\pi x e^{-\pi x^2},$$

we may rewrite this as

$$F'(\xi) = i \int_{-\infty}^{\infty} \frac{d}{dx}(e^{-\pi x^2})\, e^{-2\pi i x \xi}\, dx.$$

Integrating by parts, the boundary term vanishes because $e^{-\pi x^2} \to 0$ as $|x| \to \infty$. Hence

$$F'(\xi) = -i \int_{-\infty}^{\infty} e^{-\pi x^2} \frac{d}{dx}(e^{-2\pi i x \xi})\, dx = -i \int_{-\infty}^{\infty} e^{-\pi x^2} (-2\pi i \xi) e^{-2\pi i x \xi}\, dx.$$

Thus

$$F'(\xi) = -2\pi \xi\, F(\xi).$$

Therefore F solves

$$F'(\xi) = -2\pi \xi F(\xi),$$

hence

$$F(\xi) = Ce^{-\pi\xi^2}.$$

Evaluate at $\xi = 0$:

$$F(0) = \int_{-\infty}^{\infty} e^{-\pi x^2}\, dx = 1.$$

So $C = 1$, and therefore

$$\widehat{g}(\xi) = e^{-\pi\xi^2}.$$

□

2.2 Theta transformation

Lemma 2.2 (Theta transformation)**.** *Define*

$$\theta(t) := \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}, \qquad t > 0.$$

Then

$$\theta(t) = t^{-1/2}\theta(1/t).$$

Proof. Apply Poisson summation to

$$f_t(x) := e^{-\pi t x^2}.$$

By Gaussian self-duality with scaling,

$$\widehat{f}_t(\xi) = t^{-1/2}e^{-\pi\xi^2/t}.$$

Hence

$$\sum_{n\in\mathbb{Z}} e^{-\pi t n^2} = \sum_{m\in\mathbb{Z}} \widehat{f}_t(m) = t^{-1/2} \sum_{m\in\mathbb{Z}} e^{-\pi m^2/t}.$$

This is exactly

$$\theta(t) = t^{-1/2}\theta(1/t).$$

□

2.3 Functional equation

Theorem 2.3 (Functional equation of the completed zeta function)**.** *Define*

$$\xi(s) := \pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s).$$

Then ξ extends meromorphically and satisfies

$$\xi(s) = \xi(1-s).$$

Proof. For $\Re(s) > 1$,

$$\theta(t) - 1 = 2\sum_{n=1}^{\infty} e^{-\pi n^2 t}.$$

Multiply by $t^{s/2-1}$ and integrate on $(0,\infty)$:

$$\int_0^{\infty} (\theta(t) - 1)t^{s/2-1}\, dt = 2\sum_{n=1}^{\infty} \int_0^{\infty} e^{-\pi n^2 t} t^{s/2-1}\, dt.$$

Substitute $u = \pi n^2 t$:

$$\int_0^\infty e^{-\pi n^2 t} t^{s/2-1}\, dt = (\pi n^2)^{-s/2}\Gamma\left(\frac{s}{2}\right).$$

Therefore

$$\int_0^\infty (\theta(t)-1)t^{s/2-1}\, dt = 2\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s),$$

hence

$$\xi(s) = \frac{1}{2}\int_0^\infty (\theta(t)-1)t^{s/2-1}\, dt.$$

Split at 1:

$$\xi(s) = \frac{1}{2}\int_0^1 (\theta(t)-1)t^{s/2-1}\, dt + \frac{1}{2}\int_1^\infty (\theta(t)-1)t^{s/2-1}\, dt.$$

On $(0,1)$, use

$$\theta(t) = t^{-1/2}\theta(1/t)$$

and substitute $t = 1/u$. After the standard rearrangement, the transformed integral becomes the same expression with s replaced by $1-s$. Hence

$$\xi(s) = \xi(1-s).$$

By analytic continuation, the identity extends meromorphically to all $s \in \mathbb{C}$. □

Corollary 2.4 (Symmetry of nontrivial zeros)**.** *If*

$$\xi(\rho) = 0,$$

then

$$\xi(1-\rho) = 0.$$

Hence the nontrivial zeros are symmetric with respect to the critical line

$$\Re(s) = \frac{1}{2}.$$

Proof. Immediate from the functional equation. □

3 Core objects of the excess framework

Definition 3.1 (Explicit residue distribution)**.** Let μ denote the explicit-formula residue distribution on $\mathbb{R}$ in a Guinand–Weil normalization.

Definition 3.2 (Dipole-regularized residue channel)**.** Fix a Schwartz atom $\psi \in \mathcal{S}(\mathbb{R})$ satisfying the dipole condition

$$\widehat{\psi}(0) = 0,$$

and define

$$f := \psi * \mu.$$

Definition 3.3 (Witness bank)**.** Fix dyadic scales

$$a_j = 2^j a_0, \qquad j = 0, \ldots, J,$$

and positive weights $w_j > 0$. Define the second-difference witness response by

$$(W_a h)(\tau) := h(\tau + a) - 2h(\tau) + h(\tau - a).$$

Let $\varphi_{\tau,L}$ be a nonnegative Schwartz window localized on $[\tau, \tau + L]$. Define the bank energy

$$Q_{\text{bank}}(f;\tau,L) := \sum_{j=0}^{J} w_j^2 \,\langle W_{a_j} f, \varphi_{\tau,L} W_{a_j} f\rangle.$$

Definition 3.4 (Excess transport density)**.** Let $E_0 \geq 0$ be the baseline density. Define

$$E_{\mathrm{ex}}(f) := \limsup_{L\to\infty} \sup_{\tau\in\mathbb{R}} \frac{Q_{\mathrm{bank}}(f;\tau,L) - E_0 L}{L}.$$

4 Closure side

Definition 4.1 (Exact window law)**.** We say that f satisfies the exact window law if there exist

- a bounded function $J_f : \mathbb{R} \to \mathbb{R}$,
- a remainder $R(\tau, L)$,

such that

$$Q_{\mathrm{bank}}(f;\tau,L) = E_0 L + J_f(\tau + L) - J_f(\tau) + R(\tau, L),$$

with

$$\sup_{\tau\in\mathbb{R}} |J_f(\tau)| < \infty, \qquad \lim_{L\to\infty} \sup_{\tau\in\mathbb{R}} \frac{|R(\tau,L)|}{L} = 0.$$

Theorem 4.2 (Vanishing excess under the exact window law)**.** *If f satisfies the exact window law, then*

$$E_{\mathrm{ex}}(f) = 0.$$

Proof. Divide the exact window law by L:

$$\frac{Q_{\mathrm{bank}}(f;\tau,L) - E_0 L}{L} = \frac{J_f(\tau + L) - J_f(\tau)}{L} + \frac{R(\tau,L)}{L}.$$

Take absolute values and then the supremum over τ:

$$\sup_{\tau\in\mathbb{R}} \left| \frac{Q_{\mathrm{bank}}(f;\tau,L) - E_0 L}{L} \right| \leq \sup_{\tau\in\mathbb{R}} \left| \frac{J_f(\tau + L) - J_f(\tau)}{L} \right| + \sup_{\tau\in\mathbb{R}} \frac{|R(\tau,L)|}{L}.$$

Since J_f is bounded,

$$\sup_{\tau\in\mathbb{R}} \left| \frac{J_f(\tau + L) - J_f(\tau)}{L} \right| \leq \frac{2 \sup_\tau |J_f(\tau)|}{L} \to 0.$$

By assumption,

$$\sup_{\tau\in\mathbb{R}} \frac{|R(\tau,L)|}{L} \to 0.$$

Therefore

$$\limsup_{L\to\infty} \sup_{\tau\in\mathbb{R}} \frac{Q_{\mathrm{bank}}(f;\tau,L) - E_0 L}{L} = 0.$$

Hence

$$E_{\mathrm{ex}}(f) = 0.$$

□

5 Off-critical drift and quartic penalty

Definition 5.1 (Lifted drift)**.** Let

$$\rho = \beta + i\gamma$$

be a nontrivial zero of ζ. In log-time $\tau = \log t$, define the lifted drift

$$\delta_\rho(\tau) := \frac{\beta - \frac{1}{2}}{2\pi}\tau + r_\rho(\tau),$$

where r_ρ is bounded.

Definition 5.2 (Quartic drift penalty)**.** Define

$$P_4(\rho; \tau, L) := \int_\tau^{\tau+L} |\delta_\rho(u)|^4 \, du.$$

Lemma 5.3 (Coercive quartic growth)**.** *Assume $\beta \neq \frac{1}{2}$ and r_ρ is bounded. Then there exist constants $c, C > 0$ such that for all sufficiently large τ and all $L \geq 1$,*

$$P_4(\rho; \tau, L) \geq c \left|\beta - \frac{1}{2}\right|^4 \tau^4 L - C\tau^3 L.$$

Proof. Write

$$\delta_\rho(u) = au + b(u), \qquad a := \frac{\beta - \frac{1}{2}}{2\pi} \neq 0,$$

with $b(u) := r_\rho(u)$ bounded. There exists $M > 0$ such that

$$|b(u)| \leq M \qquad \forall u.$$

Hence

$$|\delta_\rho(u)| = |au + b(u)| \geq |a|u - M.$$

For sufficiently large u, the right-hand side is positive, and

$$|\delta_\rho(u)|^4 \geq (|a|u - M)^4.$$

Expanding the quartic polynomial gives

$$(|a|u - M)^4 = |a|^4 u^4 - 4|a|^3 M u^3 + 6|a|^2 M^2 u^2 - 4|a| M^3 u + M^4.$$

Thus, for sufficiently large u,

$$|\delta_\rho(u)|^4 \geq c_1 |a|^4 u^4 - C_1 u^3$$

for suitable constants $c_1, C_1 > 0$. Integrating over $[\tau, \tau + L]$,

$$P_4(\rho; \tau, L) \geq c_1 |a|^4 \int_\tau^{\tau+L} u^4 \, du - C_1 \int_\tau^{\tau+L} u^3 \, du.$$

For $u \in [\tau, \tau + L]$ we have $u \geq \tau$, so

$$\int_\tau^{\tau+L} u^4 \, du \geq \tau^4 L.$$

Also, for $L \geq 1$ and large τ,

$$\int_\tau^{\tau+L} u^3 \, du \leq C_2 \tau^3 L$$

for some constant $C_2 > 0$. Hence

$$P_4(\rho; \tau, L) \geq c|a|^4 \tau^4 L - C\tau^3 L.$$

Since $|a| = \frac{1}{2\pi}\left|\beta - \frac{1}{2}\right|$, this gives

$$P_4(\rho; \tau, L) \geq c \left|\beta - \frac{1}{2}\right|^4 \tau^4 L - C\tau^3 L.$$

□

6 Bridge inequality and positive excess

Proposition 6.1 (Target bridge inequality)**.** *Let $\rho = \beta + i\gamma$ be an off-critical zero with $\beta \neq \frac{1}{2}$. A sufficient bridge for positive excess is the existence of constants $c_1 > 0$, $c_2 \geq 0$ such that*

$$Q_{\text{bank}}(f; \tau, L) - E_0 L \geq c_1 P_4(\rho; \tau, L) - c_2 L$$

for all sufficiently large τ and all $L \geq 1$.

Theorem 6.2 (Conditional positive excess)**.** *If the target bridge inequality holds for some off-critical zero ρ, then*

$$E_{\text{ex}}(f) > 0.$$

Proof. Divide the bridge inequality by L:

$$\frac{Q_{\text{bank}}(f; \tau, L) - E_0 L}{L} \geq c_1 \frac{P_4(\rho; \tau, L)}{L} - c_2.$$

By the coercive quartic growth lemma,

$$\frac{P_4(\rho; \tau, L)}{L} \geq c \left|\beta - \frac{1}{2}\right|^4 \tau^4 - C\tau^3$$

for sufficiently large τ. Hence

$$\frac{Q_{\text{bank}}(f; \tau, L) - E_0 L}{L} \geq c_1 \Big(c \left|\beta - \frac{1}{2}\right|^4 \tau^4 - C\tau^3 \Big) - c_2.$$

The right-hand side tends to $+\infty$ as $\tau \to \infty$. Therefore for sufficiently large τ, it is strictly positive. Taking $\sup_{\tau \in \mathbb{R}}$ and then $\limsup_{L \to \infty}$, we obtain

$$E_{\text{ex}}(f) > 0.$$

□

Remark 6.3. A stronger version yields $E_{\text{ex}}(f) = +\infty$ if one uses the full polynomial lower bound coming from

$$\int_{\tau}^{\tau+L} u^4 \, du = \tau^4 L + 2\tau^3 L^2 + 2\tau^2 L^3 + \tau L^4 + \frac{1}{5} L^5.$$

For the contradiction principle, however, strict positivity already suffices.

7 Conditional RH theorem

Theorem 7.1 (RH from closure plus bridge)**.** *Assume:*

(i) *the exact window law holds for f, hence $E_{\text{ex}}(f) = 0$;*

(ii) *the target bridge inequality holds for every off-critical zero.*

Then every nontrivial zero of $\zeta(s)$ satisfies

$$\Re(s) = \frac{1}{2}.$$

Proof. By the vanishing excess theorem,

$$E_{\mathrm{ex}}(f) = 0.$$

Assume for contradiction that there exists an off-critical zero

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2}.$$

Then by the conditional positive excess theorem,

$$E_{\mathrm{ex}}(f) > 0.$$

This contradicts $E_{\mathrm{ex}}(f) = 0$. Therefore no off-critical zero exists. Hence every nontrivial zero lies on the critical line:

$$\Re(\rho) = \frac{1}{2}.$$

□

8 What is established and what is still missing

8.1 Established spine

The proof spine currently established is:

1. Gaussian self-duality:
$$e^{-\pi x^2} \longmapsto e^{-\pi \xi^2}.$$

2. Theta transformation:
$$\theta(t) = t^{-1/2}\theta(1/t).$$

3. Functional equation:
$$\xi(s) = \xi(1-s).$$

4. Exact window law:
$$E_{\mathrm{ex}}(f) = 0.$$

5. Off-critical drift with quartic coercive growth:
$$\beta \neq \frac{1}{2} \Longrightarrow P_4(\rho; \tau, L) \text{ grows coercively.}$$

6. Target bridge inequality:
$$\beta \neq \frac{1}{2} \Longrightarrow E_{\mathrm{ex}}(f) > 0.$$

7. Contradiction principle:
$$E_{\mathrm{ex}}(f) = 0 \quad \text{and} \quad E_{\mathrm{ex}}(f) > 0 \quad \Rightarrow \quad \text{impossible.}$$

8.2 Exact missing tasks

To upgrade the conditional theorem to a fully unconditional theorem, the following items remain to be proved without hidden assumptions or circularity:

M1. Explicit spectral extraction. Write the off-critical contribution f_ρ in a form compatible with the chosen residue channel and dipole regularization.

M2. Transport formula. Derive the lifted drift law

$$\delta_\rho(\tau) = \frac{\beta - \frac{1}{2}}{2\pi}\tau + r_\rho(\tau)$$

with bounded remainder from the explicit off-critical component.

M3. Quantitative witness lower bound. Show that the finite witness bank detects the off-critical mode with a lower bound proportional to a coercive functional of the lifted drift.

M4. No-cancellation / no-hiding estimate. Show that the remainder channel cannot cancel the coercive contribution at the level required for the target bridge inequality.

M5. Uniformity in (τ, L). Verify that the estimates hold uniformly in the regime required by the definition of $E_{\mathrm{ex}}(f)$.

9 Final verdict

Theorem 9.1 (Current exact logical status)**.** *Within the present framework, the Riemann Hypothesis is reduced to the unconditional proof of the transport-to-bank bridge, i.e. to the unconditional verification of the missing tasks **M1–M5**.*

Equivalently, the current framework yields a rigorous conditional theorem:

$$\textit{Exact window law} \quad + \quad \textit{Target bridge inequality for every off-critical zero} \quad \Longrightarrow \quad \textit{RH.}$$

Proof. Everything before this point has already been proved in the previous sections. The only nontrivial step not yet promoted to unconditional status is the bridge from an off-critical spectral component to the required bank-level lower bound. Hence the entire remaining burden is precisely the set of missing tasks **M1–M5**. □

Remark 9.2. Compressed into one line:

$$\boxed{\text{closure is in place, obstruction is in place, the missing step is the transport-to-bank bridge.}}$$

Remark 9.3. Even shorter:

$$\boxed{\text{off-critical zero} \Longrightarrow \text{linear drift} \Longrightarrow \text{quartic penalty} \Longrightarrow E_{\mathrm{ex}}(f) > 0,}$$

and the only open issue is proving this chain unconditionally from the explicit residue channel.

10 Addendum B: transport of an off-critical component to lifted drift

10.1 Purpose

The goal of this addendum is to isolate the precise transport step

$$\text{off-critical zero } \rho = \beta + i\gamma,\ \beta \neq \frac{1}{2} \quad \Longrightarrow \quad \delta_\rho(\tau) = a\tau + r_\rho(\tau),$$

with

$$a = \frac{\beta - \frac{1}{2}}{2\pi} \neq 0, \qquad r_\rho \text{ bounded.}$$

10.2 Standing notation

Let

$$f = \psi * \mu$$

be the dipole-regularized residue channel, where

$$\widehat{\psi}(0) = 0.$$

Fix a nontrivial zero

$$\rho = \beta + i\gamma.$$

Let f_ρ denote the explicit single-zero contribution of ρ to the residue channel.

Definition 10.1 (Single-zero transported channel)**.** For the selected witness observable T, define the transported single-zero signal

$$\mathcal{T}_\rho(\tau) := T(f_\rho)(\tau).$$

Its phase-unwrapped readout is denoted

$$\Phi_\rho(\tau) := \operatorname{UnwrapArg} \mathcal{T}_\rho(\tau),$$

whenever the phase readout is defined.

Definition 10.2 (Lifted drift)**.** Define the lifted drift by

$$\delta_\rho(\tau) := \Phi_\rho(\tau) - \Phi_{\text{crit}}(\tau),$$

where Φ_{crit} is the corresponding critical-line reference phase.

Proposition 10.3 (Target transport law)**.** *Assume the single-zero channel f_ρ is normalized so that the off-critical contribution is isolated in the witness readout. Then the lifted drift has the form*

$$\delta_\rho(\tau) = a\tau + r_\rho(\tau), \qquad a = \frac{\beta - \frac{1}{2}}{2\pi},$$

where r_ρ is bounded on $[\tau_0, \infty)$ for some τ_0.

Proof skeleton. The proof should be organized into the following steps.

Step B1: explicit formula extraction. Write the single-zero contribution f_ρ in the chosen normalization. This should produce a term of the form

$$f_\rho(\tau) \sim A_\rho(\tau) e^{i\gamma\tau}$$

with an amplitude modulation depending on $\beta - \frac{1}{2}$.

Step B2: dipole regularization compatibility. Show that convolution with ψ preserves the off-critical phase drift while removing the zero-frequency obstruction. More precisely, prove that the chosen normalization does not destroy the linear-in-τ phase bias induced by $\beta - \frac{1}{2}$.

Step B3: phase unwrapping. Define the phase continuously on the large-τ regime and show that the unwrapped phase difference between the off-critical mode and the critical reference satisfies

$$\Phi_\rho(\tau) - \Phi_{\text{crit}}(\tau) = a\tau + r_\rho(\tau).$$

Step B4: bounded remainder. Prove that all residual oscillatory terms remain bounded after subtraction of the linear drift. This is the precise place where one needs a clean large-τ asymptotic estimate adapted to the witness normalization.

Once Steps B1–B4 are completed, the proposition follows. □

Remark 10.4. This addendum isolates the exact content of task **M2**: deriving the linear lifted drift with bounded remainder from the explicit off-critical channel.

11 Addendum C: witness lower bound from the transported drift

11.1 Purpose

The goal of this addendum is to show that once the transported drift law

$$\delta_\rho(\tau) = a\tau + r_\rho(\tau), \qquad a \neq 0,$$

is available, the witness bank produces a coercive lower bound.

11.2 Standing notation

Let

$$P_4(\rho; \tau, L) := \int_\tau^{\tau+L} |\delta_\rho(u)|^4 \, du.$$

Recall the bank energy

$$Q_{\text{bank}}(f; \tau, L).$$

Lemma 11.1 (Coercive quartic growth)**.** *Assume*

$$\delta_\rho(u) = au + b(u), \qquad a \neq 0, \qquad |b(u)| \leq M \text{ for all } u \geq \tau_0.$$

Then there exist constants $c, C > 0$ such that for all sufficiently large τ and all $L \geq 1$,

$$P_4(\rho; \tau, L) \geq c|a|^4\tau^4 L - C\tau^3 L.$$

Proof. Since b is bounded,

$$|\delta_\rho(u)| = |au + b(u)| \geq |a|u - M.$$

Hence for large u,

$$|\delta_\rho(u)|^4 \geq (|a|u - M)^4.$$

Expand:

$$(|a|u - M)^4 = |a|^4u^4 - 4|a|^3Mu^3 + 6|a|^2M^2u^2 - 4|a|M^3u + M^4.$$

Therefore

$$|\delta_\rho(u)|^4 \geq c_1|a|^4u^4 - C_1u^3$$

for sufficiently large u. Integrating on $[\tau, \tau + L]$ gives

$$P_4(\rho; \tau, L) \geq c|a|^4\tau^4L - C\tau^3L.$$

□

Proposition 11.2 (Target witness lower bound)**.** *Assume the witness bank is normalized so that the off-critical single-zero channel contributes monotonically to the selected bank functional. Then there exist constants $c_1 > 0$, $c_2 \geq 0$ such that*

$$Q_{\text{bank}}(f; \tau, L) - E_0L \geq c_1P_4(\rho; \tau, L) - c_2L$$

for all sufficiently large τ and all $L \geq 1$.

Proof skeleton. The proof should be decomposed into the following substeps.

Step C1: identify the witness observable. Choose the concrete witness family W_{a_j} and write the single-zero response explicitly.

Step C2: isolate the drift-sensitive part. Show that the bank energy contains a positive component that depends on the lifted drift δ_ρ, and that this component controls

$$P_4(\rho; \tau, L).$$

Step C3: pass from pointwise drift to integrated bank response. Prove that after windowing and summing over the frozen finite bank, the contribution of the off-critical mode is bounded below by

$$c_1P_4(\rho; \tau, L) - c_2L.$$

Step C4: verify uniformity in the admissible regime. Check that the constants are independent of τ and L in the regime required by the definition of $E_{\text{ex}}(f)$.

This yields the target bridge inequality. □

Corollary 11.3 (Positive excess under the target witness lower bound)**.** *Assume the target witness lower bound holds and ρ is off-critical. Then*

$$E_{\text{ex}}(f) > 0.$$

Proof. Divide the bridge inequality by L:

$$\frac{Q_{\text{bank}}(f; \tau, L) - E_0L}{L} \geq c_1\frac{P_4(\rho; \tau, L)}{L} - c_2.$$

By the coercive quartic growth lemma,

$$\frac{P_4(\rho; \tau, L)}{L} \geq c|a|^4\tau^4 - C\tau^3.$$

Hence

$$\frac{Q_{\text{bank}}(f; \tau, L) - E_0L}{L} \geq c_1(c|a|^4\tau^4 - C\tau^3) - c_2.$$

For sufficiently large τ, the right-hand side is strictly positive. Taking $\sup_\tau$ and then $\limsup_{L\to\infty}$ yields

$$E_{\text{ex}}(f) > 0.$$

□

Remark 11.4. This addendum isolates the exact content of task **M3**: proving that the witness bank really detects the transported off-critical drift with a quantitative lower bound.

12 Addendum D: no-cancellation / no-hiding at bank level

12.1 Purpose

The goal of this addendum is to prove that the off-critical contribution cannot be hidden or cancelled by the remainder channel at the scale relevant for the excess functional.

12.2 Standing decomposition

Write

$$f = f_\rho + f_{\mathrm{rem}},$$

where f_ρ is the selected off-critical single-zero channel and f_{rem} is the remainder.

Let

$$Q_{\mathrm{bank}}(f;\tau,L) = Q_{\mathrm{bank}}(f_\rho;\tau,L) + Q_{\mathrm{cross}}(\rho,\mathrm{rem};\tau,L) + Q_{\mathrm{bank}}(f_{\mathrm{rem}};\tau,L)$$

be the corresponding decomposition into pure and mixed terms.

Proposition 12.1 (Target no-cancellation statement)**.** *Assume the off-critical channel f_ρ satisfies the witness lower bound of Addendum C. Assume moreover that the mixed and remainder terms satisfy*

$$Q_{\mathrm{cross}}(\rho,\mathrm{rem};\tau,L) \geq -\varepsilon\, P_4(\rho;\tau,L) - C_\varepsilon L,$$

and

$$Q_{\mathrm{bank}}(f_{\mathrm{rem}};\tau,L) \geq -C' L$$

for every $\varepsilon > 0$, all sufficiently large τ, and all $L \geq 1$.

Then for sufficiently small ε,

$$Q_{\mathrm{bank}}(f;\tau,L) - E_0 L \geq c_1' P_4(\rho;\tau,L) - c_2' L$$

for suitable constants $c_1' > 0$, $c_2' \geq 0$.

Proof. By decomposition,

$$Q_{\mathrm{bank}}(f;\tau,L) - E_0 L = \big(Q_{\mathrm{bank}}(f_\rho;\tau,L) - E_0 L\big) + Q_{\mathrm{cross}}(\rho,\mathrm{rem};\tau,L) + Q_{\mathrm{bank}}(f_{\mathrm{rem}};\tau,L).$$

Insert the lower bounds:

$$Q_{\mathrm{bank}}(f_\rho;\tau,L) - E_0 L \geq c_1 P_4(\rho;\tau,L) - c_2 L,$$

$$Q_{\mathrm{cross}}(\rho,\mathrm{rem};\tau,L) \geq -\varepsilon P_4(\rho;\tau,L) - C_\varepsilon L,$$

$$Q_{\mathrm{bank}}(f_{\mathrm{rem}};\tau,L) \geq -C' L.$$

Therefore

$$Q_{\mathrm{bank}}(f;\tau,L) - E_0 L \geq (c_1 - \varepsilon) P_4(\rho;\tau,L) - (c_2 + C_\varepsilon + C')L.$$

Choose $\varepsilon < c_1$. Then

$$c_1' := c_1 - \varepsilon > 0, \qquad c_2' := c_2 + C_\varepsilon + C'$$

give

$$Q_{\mathrm{bank}}(f;\tau,L) - E_0 L \geq c_1' P_4(\rho;\tau,L) - c_2' L.$$

□

Corollary 12.2 (No-hiding principle)**.** *Under the assumptions of the proposition, an off-critical mode cannot be hidden by the remainder channel at the scale relevant for $E_{\mathrm{ex}}(f)$.*

Proof. The proposition shows that the full bank still satisfies a bridge inequality with a positive quartic coefficient. Therefore the remainder cannot cancel the coercive off-critical contribution in the normalized excess regime. □

Proof skeleton for the analytic input. To finish this addendum rigorously, one must prove the two estimates

$$Q_{\mathrm{cross}}(\rho, \mathrm{rem}; \tau, L) \geq -\varepsilon P_4(\rho; \tau, L) - C_\varepsilon L,$$
$$Q_{\mathrm{bank}}(f_{\mathrm{rem}}; \tau, L) \geq -C' L.$$

The analytic route should be:

1. express the cross term explicitly in the witness basis;
2. exploit oscillatory mismatch between f_ρ and the remainder;
3. prove that all negative mixed contributions are sub-coercive relative to P_4;
4. show that the pure remainder channel contributes at most baseline order in L.

□

Remark 12.3. This addendum isolates the exact content of task **M4**: proving that the off-critical signal cannot hide inside the remainder when measured by the bank.

13 Combined consequence of Addenda B/C/D

Theorem 13.1 (Bridge package from B/C/D)**.** *Assume Addendum B provides the transport law*

$$\delta_\rho(\tau) = a\tau + r_\rho(\tau), \qquad a \neq 0, \quad r_\rho \text{ bounded.}$$

Assume Addendum C provides the witness lower bound

$$Q_{\mathrm{bank}}(f_\rho; \tau, L) - E_0 L \geq c_1 P_4(\rho; \tau, L) - c_2 L.$$

Assume Addendum D provides the no-cancellation transfer of this lower bound to the full channel f. Then every off-critical zero ρ forces

$$E_{\mathrm{ex}}(f) > 0.$$

Proof. Addendum B gives the linear drift with bounded remainder. Addendum C converts this drift into a coercive quartic lower bound and hence into a positive-excess statement for the isolated channel. Addendum D transfers that positivity to the full residue channel without cancellation. Therefore

$$E_{\mathrm{ex}}(f) > 0.$$

□

Remark 13.2. Compressed into one line:

$$\boxed{\text{off-critical zero} \Longrightarrow \text{transported linear drift} \Longrightarrow \text{quartic penalty} \Longrightarrow \text{bank lower bound} \Longrightarrow E_{\mathrm{ex}}(f) > 0.}$$